

MS-UNet: A Deep Learning Framework for High-Fidelity Image Dehazing

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Abstract—The visibility of many technological applications, such as autonomous driving, remote sensing, and video monitoring, is severely hampered by hazy weather conditions, which are made worse by rising global pollution. MS-UNet (Multi-Scale U-Net), a revolutionary deep learning framework for high-fidelity image dehazing, is presented in this research. Through the utilization of a multi-scale U-Net architecture, the suggested technique successfully tackles atmospheric distortions and clarifies blurry images. To improve the model's capacity to recover haze-free photos, the framework uses a multi-scale feature extraction approach that records contextual information as well as fine-grained features. By obtaining better Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) values on the SOTS dataset, quantitative assessments show that MS-UNet performs better than a number of cutting-edge dehazing methods. The framework makes a substantial addition to the fields of computer vision and image processing, as evidenced by qualitative results that further demonstrate its potential to generate aesthetically pleasing dehazed images.

Index Terms—Image Dehazing ,MS-UNet ,PSNR , SSIM

I. INTRODUCTION

The prevalence of hazy weather is rising as a result of global pollution, which presents significant challenges for daily life and technological applications. Low visibility brought on by haze makes it difficult to drive, fly, and navigate outside. Additionally, haze restricts computer vision-based technologies like autonomous automobiles, remote sensing satellite imaging, and video surveillance. In order to solve these issues, picture dehazing—which aims to lessen atmospheric distortions and restore the clarity of foggy images—has emerged as a critical area of computer vision research.

In dehazing research, the haze formation model [1] is commonly used to mathematically represent this process, as

$$I(x) = J(x)t(x) + A(1 - t(x))$$

The hazy image is represented by $I(x)$, the haze-free picture by $J(x)$, the global atmospheric light by A , and the transmission map, which shows how much unscattered light reaches the camera, by $t(x)$. Here, the transmission $t(x)$ is dependent on the haze density and scene depth. It is commonly approximated by $t(x) = e^{-\beta d(x)}$, where $d(x)$ is the scene depth and β is the atmospheric scattering coefficient. The goal of image dehazing is to reverse the scattering effects and recover $t(x)$ and A in order to estimate $J(x)$, the clear image.

Conventional picture dehazing methods estimate atmospheric properties including transmission maps and atmospheric light using image statistics and prior knowledge, then utilize the atmospheric scattering model to restore haze-free images. These techniques are limited in their ability to handle intricate haze patterns and uneven distributions, even though they show respectable performance in some situations. The Dark Channel Prior (DCP) [2], which assumes that pixels in non-sky regions have low intensity, and its improved versions with enhancements such as bilateral filtering are widely used techniques. The Color Attenuation Prior (CAP) [3], which employs a linear model, is an alternative technique for determining scene depth from color information. These methods are computationally efficient and perform well in controlled scenarios, but they struggle with complex patterns, dense fog, and uneven haze distribution. The shortcomings of conventional approaches, which frequently produce instability and artifacts due to their reliance on predetermined priors, are driving the trend toward deep learning algorithms that learn haze removal directly from data for more dependable and flexible solutions.

Deep learning techniques are used in modern approaches to get beyond the drawbacks of traditional methodologies. Deep neural networks may learn haze removal techniques from data without depending on predetermined assumptions because they are trained directly on pairs of hazy and clear images. The U-Net architecture has drawn a lot of interest among these methods since it can rebuild sharp images while preserving minute details.

Image dehazing tasks are a good fit for the U-Net model, which is built on an encoder-decoder structure with skip connections. It can capture global contextual information while preserving local details because of its hierarchical feature extraction and reconstruction capabilities. In order for U-Net to handle structural consistency during haze removal, skip connections are essential for maintaining spatial information.

The Multiscale U-Net architecture expands on U-Net's advantages by adding features to handle a range of haze densities and patterns. Because of its multiscale architecture, which integrates features at several spatial resolutions, the network can comprehend both local and global contexts at the same time. This feature enhances the capacity to manage

intricate haze patterns and irregular distributions.

When learning hierarchical and data-driven representations, the suggested Multiscale U-Net model performs better than conventional handmade features. Furthermore, it preserves computational efficiency, providing quicker processing times and better dehazing results than current techniques.

II. RELATED WORK

By examining the statistical characteristics of hazy images, methods based on previous knowledge mainly estimate medium transmittance and ambient light intensity to model the production of haze. The atmospheric scattering model is solved to reconstruct the final haze-free image. He et al.'s dark channel prior (DCP) [2] strategy makes use of the finding that, in the majority of local image patches that do not include the sky, at least one color channel shows extremely low intensity values, frequently close to zero. It is possible to successfully repair blurry photos to crisper ones by applying this past knowledge. In a similar vein, Zhu et al. presented the color attenuation prior (CAP) [3], which estimates medium transmittance by building a linear model based on local priors and forecasts depth information. Although these techniques have proven successful, they require manually calculated statistical features, which makes parallel acceleration challenging and leads to processing that takes a long time. Furthermore, the intricate process of haze creation might not be adequately captured by the manually chosen priors, which could result in limitations like color distortions or insufficient dehazing. Both subjective visual assessments and objective performance measurements are impacted by these limitations, underscoring the need for more flexible and data-driven strategies.

The performance of prior-based haze removal techniques is essentially limited by the presumptions made or the veracity of prior knowledge for a particular scene, despite the fact that these techniques have had differing degrees of success. As a result, the constraints of manually established priors sometimes result in color distortions or inadequate haze removal. Researchers are increasingly concentrating on bringing deep learning to low-level vision tasks like picture dehazing as a result of the quick development of deep learning techniques in computer vision tasks like image classification and recognition. These strategies have demonstrated notable advancements over conventional prior-based techniques.

To predict medium transmittance using deep learning, Cai et al. presented DehazeNet [6], a trainable end-to-end network. This method uses neural networks to provide more accurate estimations than prior-based methods. In a similar vein, Li et al. introduced AOD-Net [8], an integrated dehazing network that streamlines the procedure by producing haze-free images immediately without requiring the estimation of ambient light and medium transmittance independently. Designed to be a lightweight convolutional neural network, AOD-Net can be easily integrated into other models, like Faster R-CNN, to improve performance on blurry images in high-level computer vision tasks.

Instead of using models that depend on the atmospheric scattering model, Ren et al. created a gated fusion network (GFN) [4] that allows them to immediately recover clean images from foggy inputs. GFN employs a fusion-based approach in conjunction with an encoder-decoder structure. It creates three inputs for the network by applying gamma correction, contrast enhancement, and white balance to the input hazy image. Contextual information is captured by the encoder, and the final dehazed image is produced by the decoder, which uses a confidence map to estimate pixel-level contributions and successfully blend the three inputs.

When compared to earlier methods, these deep learning-based techniques have shown higher performance in terms of accuracy and efficiency, underscoring the expanding potential of data-driven solutions in image dehazing.

Motivated by the widespread use of U-Net and multi-scale processing techniques, we suggested a Multi-Scale U-Net to enhance image dehazing effects. The following are the contributions made by the suggested Multi-Scale U-Net :

- 1) Inspired by U-Net, we proposed an end-to-end Multi-Scale U-Net with an enhanced encoder-decoder structure to enhance multi-scale feature learning for image dehazing.
- 2) The Multi-Scale U-Net's encoder module efficiently extracts picture features at various scales by integrating different resolution paths through simultaneous down-sampling operations.
- 3) A multi-scale fusion block is introduced at the encoder-decoder junction to merge features from various resolutions, improving the network's capacity to simultaneously record fine and coarse details.
- 4) Like U-Net, the network uses skip connections for training stability in order to effectively transmit both low-level and high-level features while avoiding vanishing gradient problems.
- 5) Three metrics are used in the evaluation: Learned Perceptual Image Patch Similarity (LPIPS), the Peak Signal-to-Noise Ratio (PSNR), which measures the quality of reconstruction and higher values indicate better performance, and the Structural Similarity Index (SSIM), which measures the perceptual similarity between dehazed and ground truth images.
- 6) When tested on simulated hazy images, the Multi-Scale U-Net outperforms existing state-of-the-art methods in terms of image dehazing with improved SSIM, PSNR, and LPIPS.

III. PROPOSED WORK

A. Proposed Methodology

A multi-branch feature extraction and recalibration mechanism for automatic image dehazing is suggested in an architecture based on deep convolutional neural networks. The network draws inspiration from the U-Net encoder-decoder structure, which uses multi-scale convolutional operations at different granularities for each stage of encoding and decoding

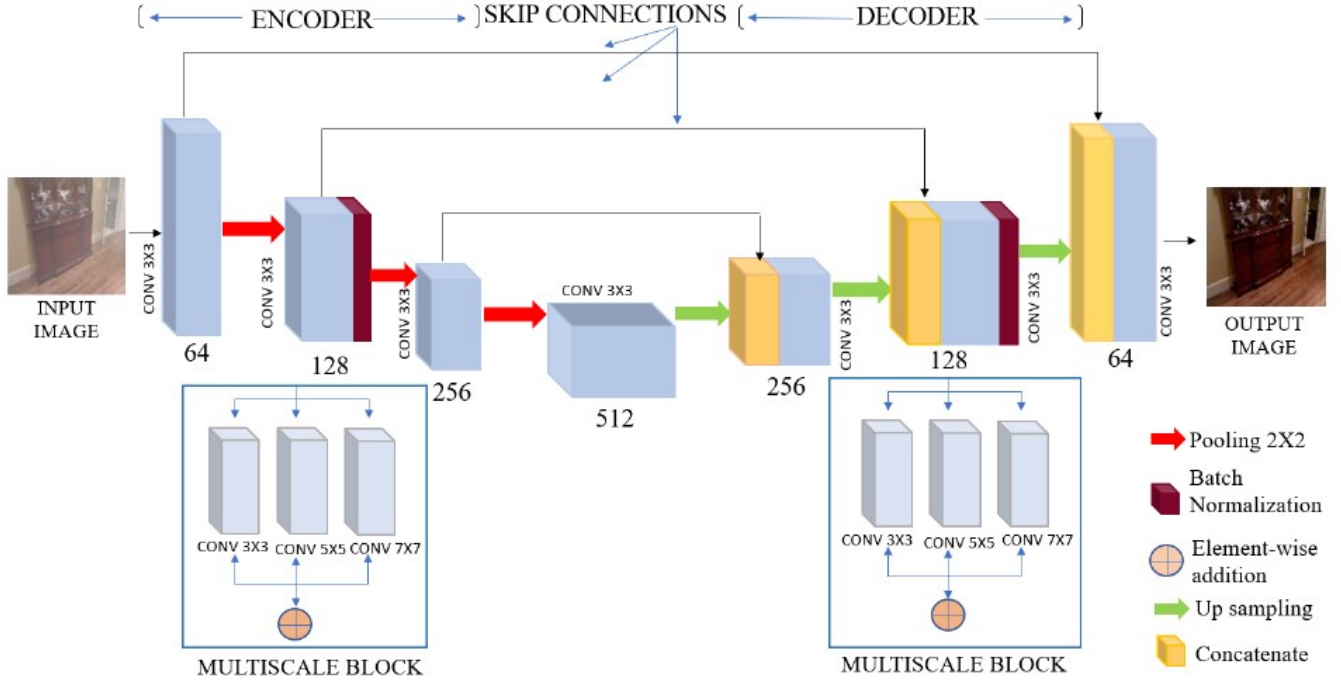


Fig. 1. Proposed MS-UNet Architecture

to extract rich spatial and contextual information. Fig. 1 shows the proposed MS-UNet Encoder-Decoder architecture. In particular, we use a multi-path feature aggregation approach that integrates three parallel convolutional branches at each stage in place of conventional convolutional layers. The model can simultaneously capture contextual features and fine-grained details since each branch runs at a different receptive field size. Element-wise addition is used to combine the outputs of various branches, guaranteeing an effective fusion of multi-scale features for enhanced dehazing performance.

To down-sample the spatial resolution while preserving important feature information, the model gradually extracts hierarchical features in the encoding path using several convolutional blocks, followed by batch normalization and max-pooling processes. The network's capacity to retain fine spatial information and boost feature representation for haze removal is improved at each level by the aggregated multi-scale features. The decoding path integrates the hierarchical multi-scale characteristics learned in the encoder and restores spatial resolution by using symmetric up-sampling and concatenation with skip connections. After every fusion, batch normalization is added to guarantee steady training and quicker convergence.

Across all convolutional blocks, we incorporate additive residual learning to improve the recalibration of both local and global features. This approach facilitates improved feature learning, enhances optimization, and lessens the vanishing gradient issue. The final output layer uses a 3x3 convolutional operation and batch normalization to create the restored



Fig. 2. Block Diagram of the Proposed Image Dehazing Methodology using MS-UNet Framework.

haze-free image. The suggested architecture improves picture dehazing by efficiently eliminating haze while maintaining edge details and fine structures by utilizing multi-scale feature extraction and fusion with residual learning.

B. Multiscale Features

The U-Net model's multi-scaling technique improves feature extraction by utilizing convolutional filters of three by three, five by five, and seven by seven sizes. The network can extract both local and global contextual information thanks to this design, which allows it to capture features at various receptive fields. An additive fusion method is used

to aggregate the outputs after the multi-scale block processes the input features through parallel convolutional layers, each of which operates at a separate scale. The network is better able to characterize features thanks to this hierarchical feature aggregation, which enhances the refinement of structural patterns and spatial details. For image dehazing tasks where texture preservation and information recovery are critical, the utilization of numerous receptive fields makes it easier to detect both fine-grained details and more general contextual associations.

By combining features at several sizes via skip connections, the multi-scale block in the decoder further improves the feature maps. By bridging the encoder and decoder stages, these skip connections guarantee the preservation of high-resolution spatial details that are lost during downsampling. BatchNormalization is applied to the multi-scale block's outputs in order to enhance convergence and stabilize learning. The network successfully overcomes the drawbacks of traditional CNNs, which frequently lose crucial contextual information as a result of pooling processes, by integrating feature maps from various scales. The final design improves the dehazing process's overall performance by more precisely reconstructing intricate textures and fine details. The suggested multi-scale block, which is based on the ideas of Res2Net, is ideal for jobs that demand for multi-level feature extraction since it exhibits enhanced feature representation through hierarchical feature splitting and merging.

The multi-scale feature aggregation process in the suggested MS U-Net model sets it apart from other dehazing techniques and improves both global contextual comprehension and fine-grained details. Our method uses parallel convolutional branches with different receptive fields (3×3, 5×5, and 7×7), which allows the network to record hazy patterns at several scales, in contrast to conventional U-Net-based models that rely on single-scale convolutions. Furthermore, effective multi-scale representation is guaranteed by the incorporation of element-wise feature fusion, and optimization and gradient flow are enhanced by additive residual learning. Compared to traditional CNN-based methods, these improvements enable MS U-Net to perform better haze removal while maintaining textures and edges.

IV. EXPERIMENT SETUP AND RESULT ANALYSIS

The effectiveness of the proposed MS-UNet in image dehazing is confirmed by comparing it with two existing classical prior knowledge-based approaches (DCP [3], CAP [4]) and four existing deep learning-based techniques (DeHazeNet [6], AOD-Net [8], and GFN [4]). The evaluation metrics for picture dehazing performance and the dataset of hazy photos utilized are initially presented in this section. Following that, it does a quantitative and qualitative evaluation of different strategies using a set of artificially created blurry images.

A. Dataset and Implementation

The SOTS [9] dataset, which is extensively used in the field of image dehazing, was used in our investigation. This dataset provides a comprehensive standard by which to measure the effectiveness of picture enhancing techniques. The 550 interior photos in SOTS are divided into 500 correspondingly synthetically hazy images and 50 original haze-free shots.

B. Performance Metrics

Evaluated the dehazing results on the SOTS dataset using two widely used metrics, PSNR (Peak Signal-to-Noise Ratio) [9], SSIM (Structural Similarity Index) [9], and LPIPS (Learned Perceptual Image Patch Similarity) [11], in order to objectively assess the performance of various image dehazing methods.

Peak Signal-to-Noise Ratio (PSNR):

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (1)$$

- Determines the proportion between the power of noise and the maximum signal power.
- $MAX \rightarrow$ Maximum possible pixel value.
- $MSE \rightarrow$ Mean Squared Error between the original and processed image.

Structural Similarity Index (SSIM):

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (2)$$

- Measures perceptual similarity between two images.
- $\mu_x, \mu_y \rightarrow$ Mean intensities of images x and y .
- $\sigma_x^2, \sigma_y^2 \rightarrow$ Variances of x and y .
- $\sigma_{xy} \rightarrow$ Covariance between x and y .
- $C_1, C_2 \rightarrow$ Small constants to avoid division by zero.

Learned Perceptual Image Patch Similarity (LPIPS):

$$LPIPS(x, \hat{x}) = \sum_l w_l \cdot \|F_l(x) - F_l(\hat{x})\|_2^2 \quad (3)$$

- Uses characteristics based on deep learning to quantify perceptual similarity.
- $x, \hat{x} \rightarrow$ Reference (haze-free) and dehazed images.
- $F_l(x), F_l(\hat{x}) \rightarrow$ Deep feature representations from layer l .
- $\|F_l(x) - F_l(\hat{x})\|_2^2 \rightarrow$ Perceptual difference in feature space.
- $w_l \rightarrow$ Layer-specific learned weights.

These measures are especially well-suited for synthetic datasets that give matched clear and hazy images since PSNR and SSIM computations require a corresponding haze-free ground truth image. On the other hand, LPIPS compares high-level deep features from a pre-trained network to quantify perceptual similarity, which enables it to assess image quality even when ground truth images are not accessible.

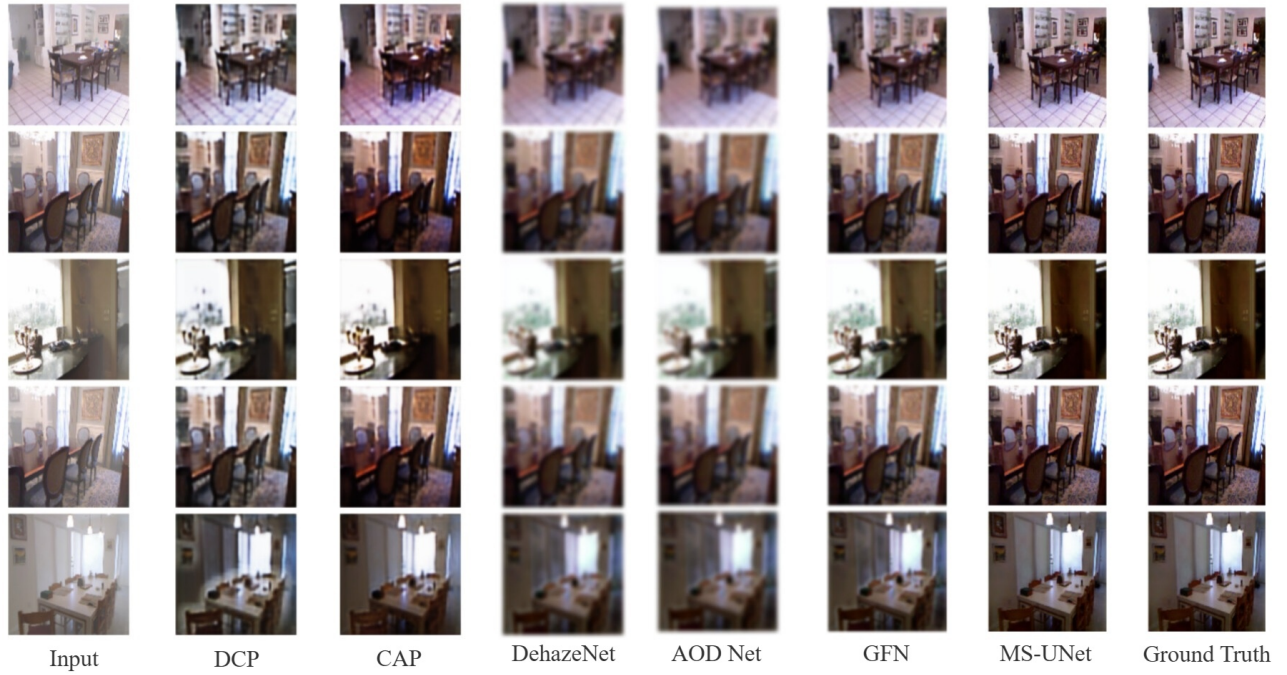


Fig. 3. Qualitative comparison of indoor synthetic hazy images from the SOTS [9] dataset

TABLE I
QUANTITATIVE COMPARISON ON AVERAGE PSNR, SSIM AND LPIPS FOR ALL THE INDOOR HAZY IMAGES FROM THE SOTS DATASET.

Method	DCP	CAP	DeHaze-Net	AOD-Net	GFN	MS-UNet
PSNR	18.87	21.31	22.06	21.01	22.99	31.67
SSIM	0.7935	0.8247	0.8325	0.8372	0.9033	0.9648
LPIPS	0.3294	0.3062	0.3103	0.2024	0.2155	0.1700

C. Implementation Parameters

Figure 1 shows the MS U-Net's full architecture and parameter configurations. In order to improve feature representation, the network employs element-wise addition after multi-scale feature extraction via concurrent $3 \times 3 \times 5 \times 7$ convolutional layers. Bilinear interpolation with skip connections for feature fusion is used to accomplish upsampling, whereas 2×2 max pooling is used for all downsampling processes. The Adam optimizer is used to train the model with an input patch size of 128×128 and a batch size of 4. With an initial learning rate of $1e-4$, the complete network is trained across 70 epochs. The loss function optimizes both pixel-wise accuracy and perceptual quality by combining Binary Crossentropy and SSIM loss. All training and testing processes are performed using Pytorch and run on a GPU.

D. Quantitative results

The quantitative assessment outcomes of several image dehazing techniques are shown in the table. The top-performing outcomes, which display the average PSNR, SSIM, and LPIPS

values attained by each technique on all indoor hazy images from the SOTS dataset, are indicated in bold. The suggested MS-UNet provides the best indoor image dehazing performance, as indicated in Table II, with the maximum PSNR of 33.04, SSIM of 0.9648, and LPIPS of 0.1700. Compared to the other six approaches, MS-UNet's two measures are substantially better. With comparatively high PSNR and SSIM values, GFN [4] performs the second-best among the reference methods. DCP [2], on the other hand, produces the lowest PSNR and SSIM values, suggesting subpar dehazing performance. With PSNR and SSIM values in the middle range, the remaining techniques—CAP [3], DeHazeNet [6], and AOD-Net [8]—produce outcomes that are moderate.

E. Qualitative results

To further subjectively evaluate the image dehazing performance of different approaches, Fig. 3 shows visual comparisons of dehazing results on several example synthetic indoor foggy images. The results of AOD-Net [8] show discernible residual haze. However, the dehazed images created by DCP

[2] exhibit observable color distortion. The outputs from CAP [3] and DeHazeNet [6] have less residual haze than AOD-Net [8], but they are still not as clear as they could be. Although GFN [4] effectively removes haze, the final image has observable artifacts, such as the uneven color distribution on the walls in the final three shots and the loss of detail in dark regions, like the shadowy corner of the building wall. On the other hand, the proposed MS-UNet does a superior job of dehazing and yields results that are extremely close to the ideal haze-free photographs, or ground truth.

V. CONCLUSION

Image dehazing has been successfully accomplished with MS-UNet, which has demonstrated remarkable outcomes in both quantitative and qualitative evaluations. The framework greatly outperformed current state-of-the-art techniques, achieving a Peak Signal-to-Noise Ratio (PSNR) of 33.04, a Structural Similarity Index (SSIM) of 0.9648, and an LPIPS of 0.1700. These outcomes demonstrate how well the model can restore information and clarity in blurry photos, which makes it a useful tool for applications that need better visibility. In addition to enhancing dehazing performance, the effective combination of multi-scale feature extraction and residual learning guarantees the preservation of edge structures and fine details. The foundation for future developments in image processing methods, especially under difficult atmospheric conditions, is laid by this work. Although multi-scale feature extraction in the suggested MS U-Net successfully improves dehazing performance, more can be done. Self-supervised learning could enhance generalization to real-world images, and adaptive attention mechanisms could aid in more efficient focus on haze-dense areas.

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